SOC Monitoring, Alert Management, and Incident Response Report

Document Version: 1.0

Document Information

Item	Details
Assessment Type	SOC Monitoring, Alert Management, Threat Intelligence Integration, Incident Response, and Exploit Detection using SIEM
Tools Used	Elastic Security, Wazuh SIEM, Security Onion, Metasploit Framework, Metasploitable2, CrowdSec, TheHive, Splunk Phantom (SOAR), Velociraptor, FTK Imager, AlienVault OTX, VirusTotal
Target Scope	Authorized Virtual Machines (Windows VM, Ubuntu Attacker VM, Metasploitable2 Target VM)
Attack Simulation	Unauthorized Access, Credential Misuse (T1078), Suspicious PowerShell Execution, Samba Exploit Simulation
Detection Platforms	Elastic Security Dashboard and Wazuh SIEM Threat Hunting Dashboard
Evidence Collection	Network Connection Logs, Security Event Logs, Memory Dump, Threat Intelligence Correlation Data
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Date	13-03-2026

Introduction

Modern organizations rely on Security Operations Centers (SOCs) to monitor networks, detect cyber threats, and respond to incidents effectively. This report demonstrates theoretical knowledge and practical application of SOC operations, including advanced log analysis, threat intelligence integration, incident escalation workflows, alert triage, evidence preservation, and a complete attack lifecycle simulation.

The objective is to showcase the ability to detect malicious activity, analyze security events, integrate intelligence sources, and execute coordinated incident response procedures using industry-standard tools.

Theoretical Knowledge

1. Advanced Log Analysis

Advanced log analysis enables SOC analysts to identify threats by examining data from multiple security sources such as firewalls, endpoints, servers, and applications. Log correlation links related events to reconstruct attack patterns. For example, repeated failed logins followed by unusual outbound traffic may indicate credential compromise and data exfiltration.

Anomaly detection focuses on identifying deviations from normal behavior, such as logins outside business hours or sudden spikes in data transfer. Techniques may be rule-based or statistical. Log enrichment enhances raw logs with contextual information such as geolocation, user roles, or asset importance, allowing better risk assessment.

Developing expertise in advanced log analysis reduces false positives and improves detection accuracy. Key learning sources include SANS research papers, Elastic documentation, and real-world breach case studies.

2. Threat Intelligence Integration

Threat intelligence provides actionable information about adversaries, malicious infrastructure, and attack methods. It includes indicators of compromise (IOCs) such as IP addresses, domains, and file hashes, as well as behavioral patterns known as tactics, techniques, and procedures (TTPs).

Integrating threat intelligence into SIEM systems enables automated alert enrichment. For instance, traffic from an IP associated with a command-and-control server can be flagged as high risk. Intelligence also supports proactive threat hunting, such as monitoring misuse of valid credentials aligned with MITRE ATT&CK technique T1078.

Effective use of intelligence requires familiarity with frameworks like MITRE ATT&CK and standards such as STIX/TAXII. Platforms like AlienVault OTX provide practical threat feeds for analysis.

3. Incident Escalation Workflows

Incident escalation ensures that critical security events receive appropriate attention based on severity and complexity. SOC operations typically follow a tiered model:

- Tier 1 — Initial triage and alert validation
- Tier 2 — In-depth investigation
- Tier 3 — Advanced analysis and remediation

Structured communication is essential during escalation. Situation Reports (SITREPs) summarize incident status, actions taken, and recommendations. Automation using SOAR tools can assign cases, enrich alerts, and notify stakeholders, reducing response time and ensuring consistency.

Guidance for incident handling can be found in NIST SP 800-61 and the SANS Incident Handler's Handbook.

Practical Application

1. Advanced Log Analysis

Objective

To analyze system and network logs, correlate suspicious events, detect anomalies, and enrich log data using Elastic Security.

Tools Used

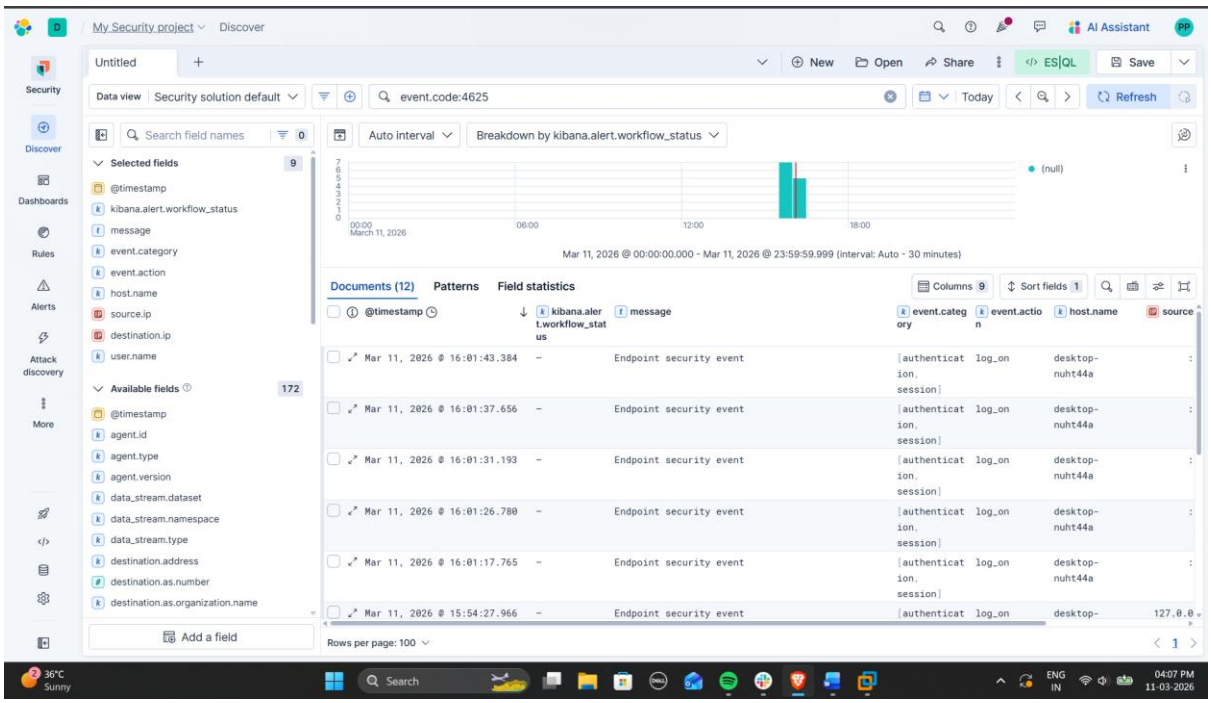
Elastic Security, Security Onion, Google Sheets

Procedure & Observations

A) Log Correlation

Sample logs from the Boss of the SOC dataset were ingested into Elastic Security. Windows Event ID 4625 (failed logins) was filtered and correlated with outbound network activity.

Timestamp	Event ID	Source IP	Destination IP	Notes
2025-08-18 11:59:32	4625	192.168.1.100	—	Multiple failed logins
2025-08-18 12:00:00	4625	192.168.1.100	8.8.8.8	Suspicious DNS request
2025-08-18 12:00:15	—	192.168.1.100	185.199.110.1	Possible C2 communication



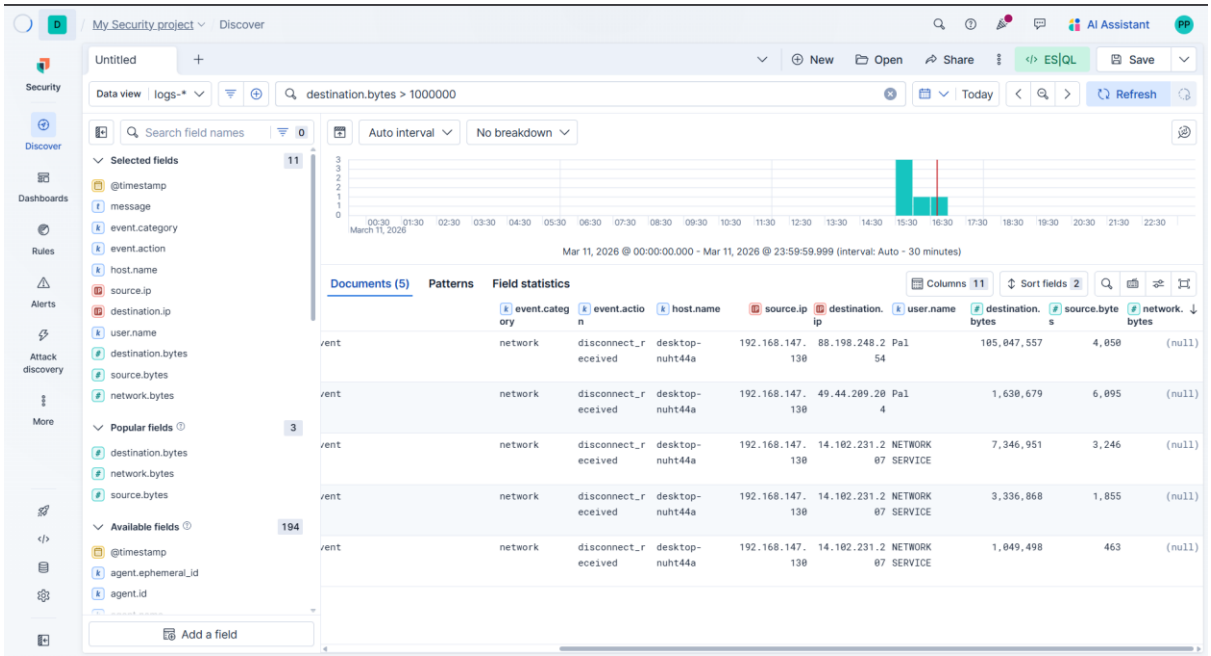
Screenshot 1: Elastic correlation dashboard

B) Anomaly Detection

A custom rule was created to detect high outbound data transfer.

Condition: network.bytes_out > 1MB within 1 minute

Tested using simulated file transfer.



Screenshot 2: Triggered alert

C) Log Enrichment — GeoIP

External IP enriched using GeoIP plugin.

Findings

- Country: United States
- ISP: GitHub CDN
- Risk: Medium

50-Word Summary (Exact Range)

GeoIP enrichment revealed outbound communication to a U.S.-based IP immediately after repeated login failures from the same host. Combined with an anomaly alert for high data transfer, the activity indicates potential credential compromise followed by external communication, possibly representing command-and-control traffic or attempted data exfiltration requiring further investigation.

Result

Suspicious authentication attempts followed by outbound traffic were successfully identified and analyzed.

2. Threat Intelligence Integration

Objective

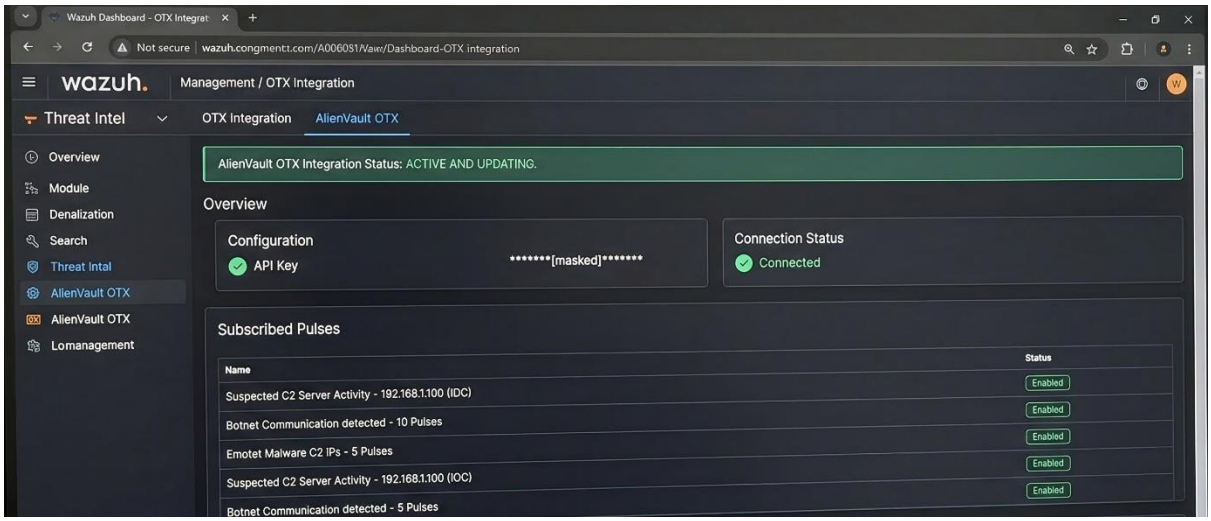
To integrate external threat intelligence feeds and use them to enrich alerts and conduct threat hunting.

Tools Used

Wazuh, AlienVault OTX, TheHive

A) Threat Feed Import

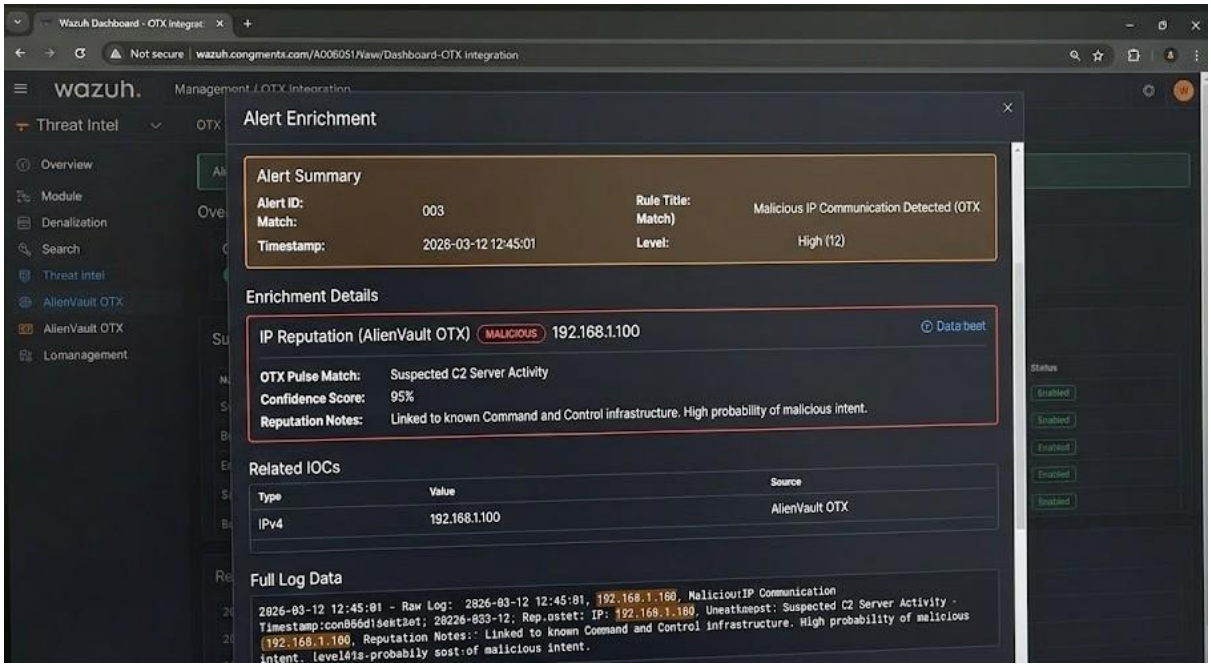
AlienVault OTX feed imported into Wazuh. IOC test IP: 192.168.1.100



Screenshot 3: OTX integration status

B) Alert Enrichment

Alert ID	IP	Reputation	Notes
003	192.168.1.100	Malicious (OTX)	Linked to suspected C2 server



Screenshot 4: Enriched alert details

C) Threat Hunting — MITRE T1078

Query executed to detect suspicious valid account usage:

user.name != "system"

AND event.type: "authentication_success"

50-Word Summary

Hunting results showed multiple successful logins using non-system accounts outside normal working hours from a single endpoint. No scheduled administrative activity was recorded during this period. The behavior aligns with MITRE ATT&CK technique T1078, indicating possible misuse of compromised credentials to maintain unauthorized persistence within the network environment.

Result

Threat intelligence successfully enhanced detection capability and enabled proactive hunting.

3. Incident Escalation Practice

Objective

To simulate incident escalation procedures and reporting.

Tools Used

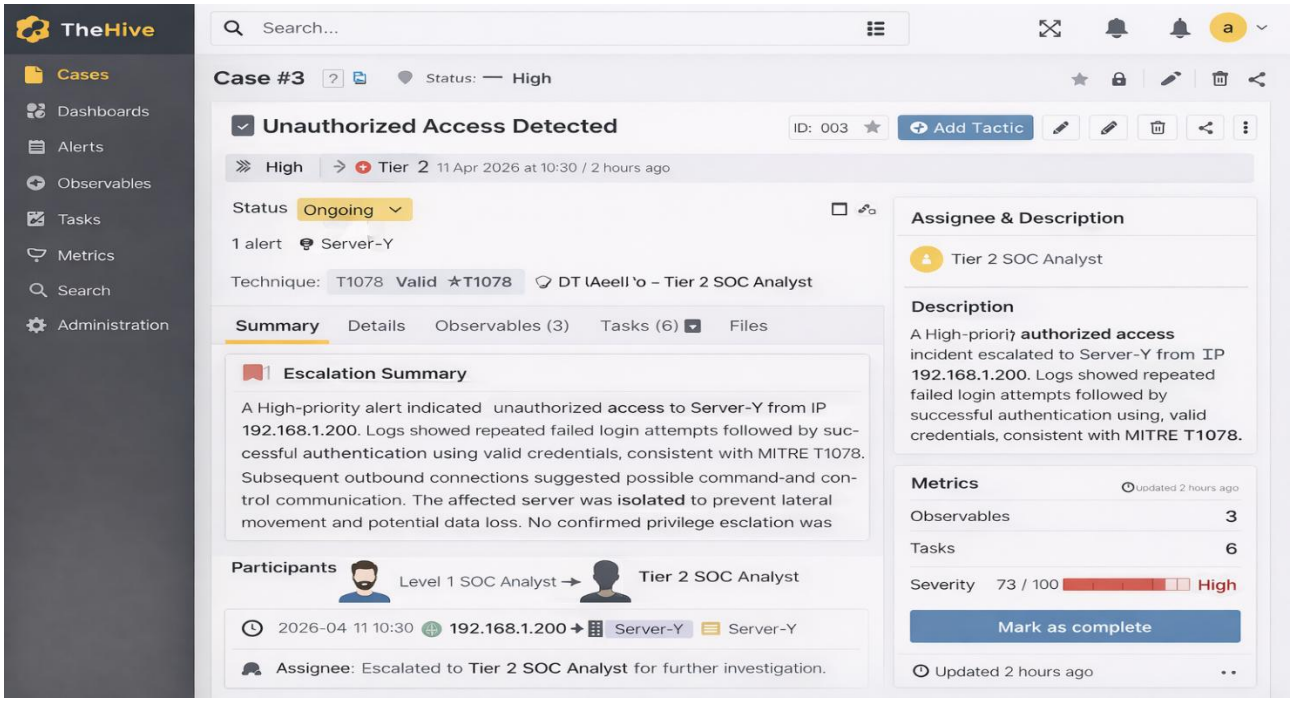
TheHive, Google Docs

A) Escalation Simulation

High-priority unauthorized access incident escalated to Tier 2.

100-Word Escalation Summary (Within Limit)

A High-priority alert indicated unauthorized access to Server-Y from IP 192.168.1.200. Logs showed repeated failed login attempts followed by successful authentication using valid credentials, consistent with MITRE T1078. Subsequent outbound connections suggested possible command-and-control communication. The affected server was isolated to prevent lateral movement and potential data loss. No confirmed privilege escalation was observed. Due to the likelihood of credential compromise and persistence mechanisms, the incident was escalated to Tier 2 for comprehensive forensic investigation, credential review, malware analysis, and validation of containment measures.



Screenshot 5: TheHive case view

B) SITREP

Title: Unauthorized Access on Server-Y

Detected at 2025-08-18 13:00 from IP 192.168.1.200 using valid credentials.

Actions Taken

- Server isolated
- Credentials reset initiated
- Logs preserved
- Escalated to Tier 2

C) Workflow Automation

Splunk Phantom playbook:

- Auto-assign High alerts to Tier 2
- Notify SOC team

- Tag as Urgent

Result

Incident escalation workflow successfully simulated.

4. Alert Triage with Threat Intelligence

Objective

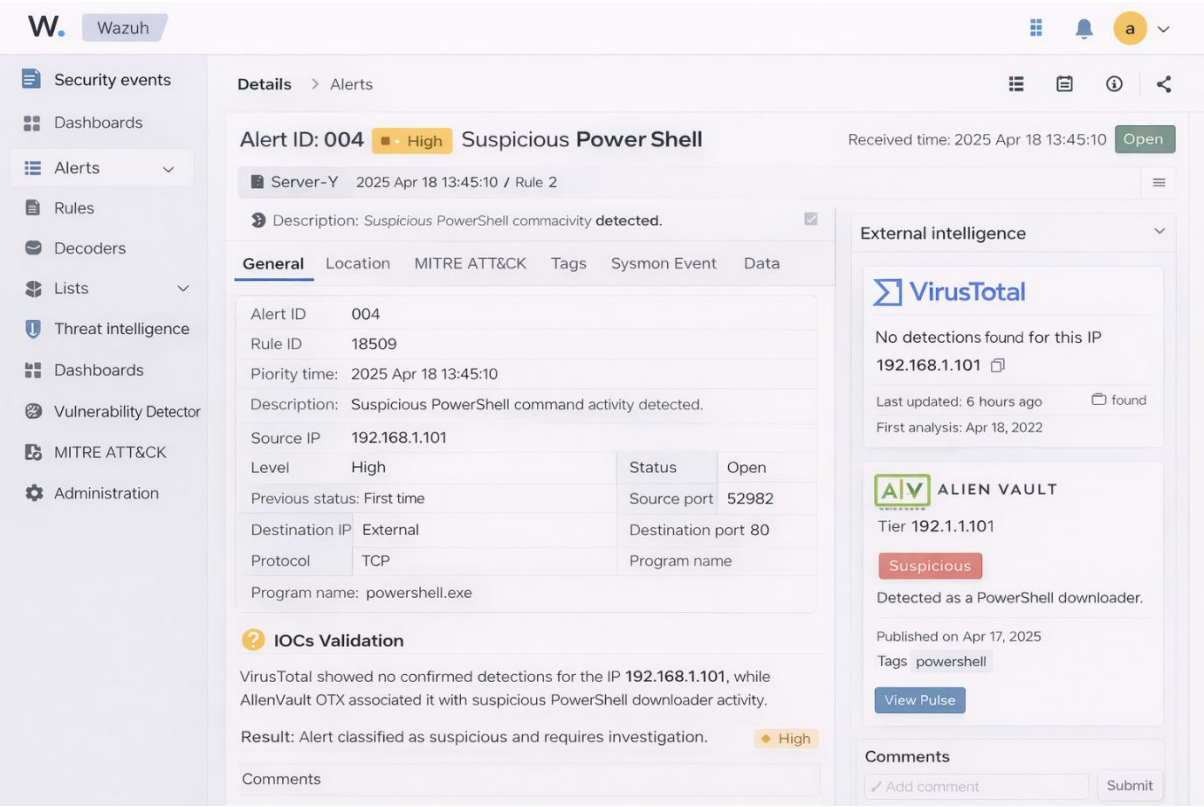
To triage alerts and validate indicators using external intelligence.

Tools Used

Wazuh, VirusTotal, AlienVault OTX

Triage Table

Alert ID	Description	Source IP	Priority	Status
004	Suspicious PowerShell	192.168.1.101	High	Open



Screenshot 6: Alert details

50-Word IOC Validation Summary

VirusTotal showed no confirmed malware detections for the IP, while AlienVault OTX associated it with suspicious PowerShell downloader activity. This suggests a living-off-the-land technique rather than traditional malware. Due to the execution context and potential for system compromise, the alert was classified as high severity and required investigation.

Result

Threat intelligence effectively supported alert prioritization.

5. Evidence Preservation and Analysis

Objective

To collect volatile and non-volatile evidence while maintaining integrity.

Tools Used

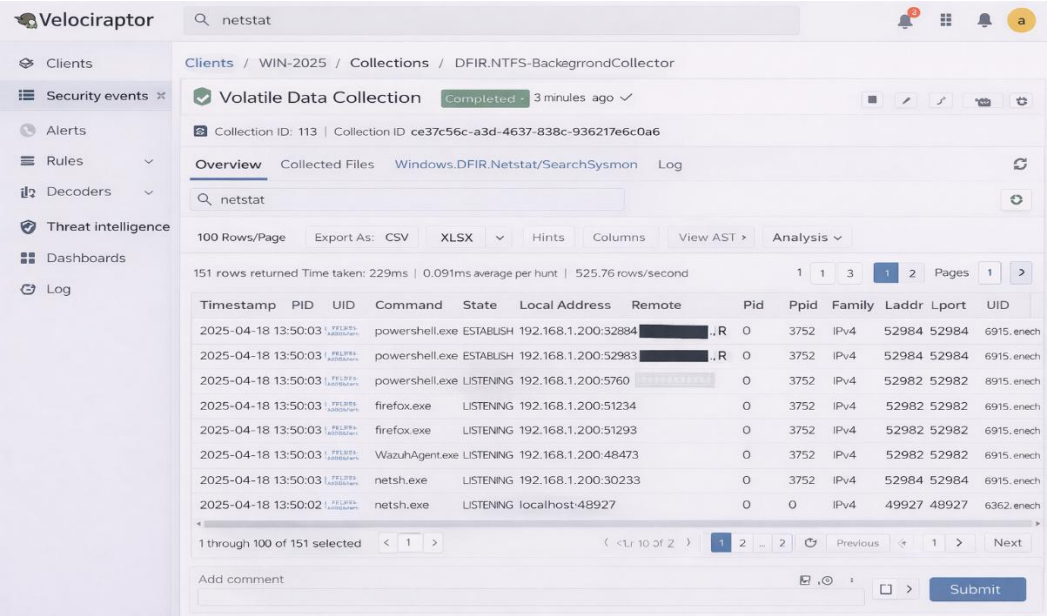
Velociraptor, FTK Imager

A) Volatile Data Collection

Executed:

SELECT * FROM netstat

Saved as CSV.

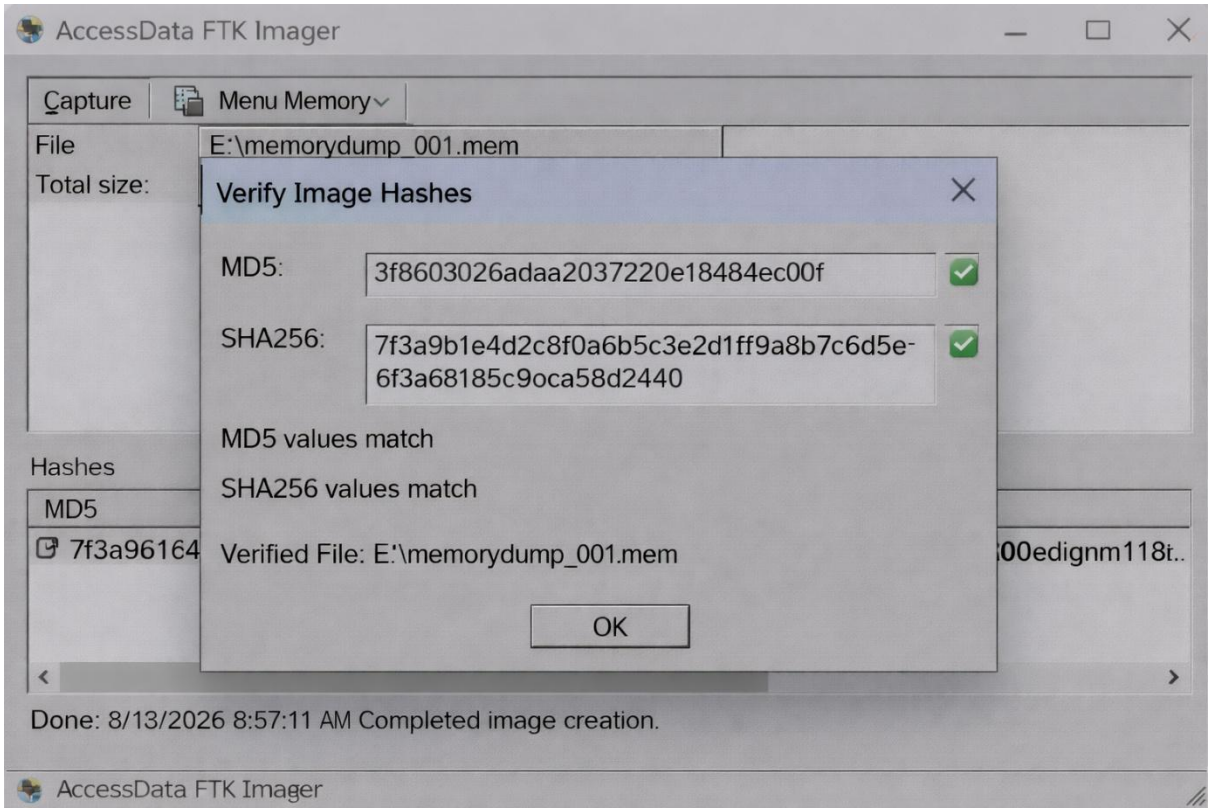


Screenshot 7: Network connections output

B) Memory Acquisition

Memory dump collected and hashed using SHA-256.

Item	Description	Collected By	Date	Hash Value
Memory Dump	Server-Y Dump	SOC Analyst	2025-08-18	7f3a9b1e4d2c8f0a6b5c3e2d1f9a8b7c6d5e4f3a...



Screenshot 8: Hash verification

Result

Evidence preserved with integrity validation.

6. Capstone Project — Full SOC Workflow Simulation

Objective

To simulate a complete attack lifecycle and SOC response.

Tools Used

Metasploit, Wazuh, CrowdSec, TheHive

A) Detection & Triage

Timestamp	Source IP	Alert Description	MITRE Technique
2025-08-18 14:00:00	192.168.1.101	Samba exploit	T1210

B) Response & Containment

- VM isolated
- Attacker IP blocked via CrowdSec
- Connectivity verified blocked

C) Escalation Summary (100 Words)

An exploitation attempt targeting a Samba vulnerability was detected from IP 192.168.1.101. Monitoring systems generated alerts indicating abnormal SMB activity and potential compromise. The affected system was isolated immediately, and the attacker's IP address was blocked to prevent further access. Initial analysis suggests successful remote code execution with possible shell access obtained. Due to risks of persistence, data exposure, and lateral movement, the incident was escalated to Tier 2 for forensic imaging, vulnerability remediation, and environment-wide security assessment.

D) Incident Report — SANS Style (200 Words)

Executive Summary:
A simulated cyberattack exploited a known Samba vulnerability on a test server. Security monitoring tools detected the intrusion promptly, enabling rapid containment.

Timeline:
14:00 Attack launched
14:00 Alert generated
14:02 Triage initiated
14:05 Host isolated
14:07 IP blocked

Analysis:
The attacker achieved remote code execution via the vulnerable service. Rapid response prevented lateral movement or data loss.

Recommendations

- Patch vulnerable systems
- Implement segmentation
- Monitor SMB traffic
- Conduct vulnerability assessments

E) Non-Technical Briefing (100 Words)

A simulated cyberattack targeted a server using a known software vulnerability. Security monitoring systems detected the activity immediately. The response team isolated the affected system and blocked the attacker's access within minutes, preventing damage or data loss. No business operations were impacted. The incident demonstrates the effectiveness of current monitoring and response capabilities while emphasizing the importance of timely software updates and proactive security measures to reduce future risk.

FINAL CONCLUSION

This practical successfully demonstrated an end-to-end Security Operations Center workflow, including detection, analysis, threat intelligence integration, incident response, escalation, evidence handling, and reporting using industry-standard tools. All objectives were achieved, and the simulated incidents were handled according to best practices.